

PART 10 OF 18

Reasoning and Planning Governance

Intent Verification, Plan Validation, Goal Consistency, Constitutional Constraints, Reasoning Anomaly Detection, and Privacy-Preserving Oversight

ENTERPRISE AI CONTROL ARCHITECTURE

Implementation Guide for Production AI Systems • 2026

10.1 The Reasoning Governance Challenge

Reasoning governance is the most technically challenging domain in enterprise AI control. Unlike action governance (which operates on discrete, observable tool calls), reasoning governance must evaluate the agent's internal cognitive process—a process that is partially opaque even to the model itself. The challenge is compounded by the need to maintain oversight without exposing sensitive chain-of-thought content that may include confidential business context.

Core Tension: Effective reasoning governance requires visibility into the agent's planning and decision-making process. However, the reasoning trace often contains sensitive information (financial analysis, legal reasoning, personal data). Governance must operate without creating additional data exposure risks.

10.2 Intent Verification

10.2.1 What Intent Verification Addresses

Intent verification is the process of evaluating whether an agent's declared goal (what it says it is trying to do) is consistent with its observed actions (what it is actually doing). Discrepancy between declared intent and action patterns is a strong signal of either misalignment or adversarial manipulation.

Verification Method	Inputs	Detects	Implementation
Goal-Action Consistency	Stated goal + action history	Actions inconsistent with goal	Semantic similarity model
Action Sequence Analysis	Ordered action history	Plans inconsistent with declared tasks	Sequential pattern classifier
Counterfactual Testing	Agent actions vs. alternative optimal paths	Suboptimal or suspicious action selection	Reference policy comparison
Information Flow Analysis	Data accessed vs. data required for goal	Unauthorized data access patterns	Data dependency graph analysis
Side-Effect Detection	Actions + environmental state changes	Unauthorized environmental modifications	State diff monitoring

10.3 Plan Validation Architecture

10.3.1 Multi-Step Plan Review

For agents with long-horizon planning capabilities, plan validation reviews the agent's intended action sequence before execution begins. This enables approval-gate efficiency: review the plan once rather than each individual action. However, plan validation only catches risks visible in the plan—emergent risks from

plan-environment interaction require runtime monitoring.

- **Plan Submission:** Agent externalizes its intended action sequence in a structured format (JSON plan object) before beginning execution
- **Plan Safety Check:** Policy engine evaluates each planned action and the action sequence for: individual action risks, sequence-level risks (e.g., does the sequence constitute data exfiltration?), and constraint violations
- **Plan Approval:** Human or automated reviewer approves, modifies, or rejects the plan before execution begins
- **Execution Monitoring:** During execution, agent's actual actions compared against approved plan; deviations trigger alerts
- **Plan Modification Protocol:** If agent needs to deviate from approved plan due to environmental changes, it must submit a plan amendment for review

10.4 Constitutional Constraints in Production

10.4.1 Constitutional AI for Enterprise

Constitutional AI (Anthropic, 2022) trains models to evaluate their own outputs against a set of principles ('constitution'). For enterprise deployment, this concept is extended to runtime enforcement: a constitution of enterprise-specific rules is encoded in the agent's system prompt and enforced through both model-level training and runtime policy checks.

10.4.2 Enterprise Constitution Elements

Constraint Category	Example Rules	Enforcement Layer
Data Handling	Never transmit PII to external services; mask sensitive data in logs	System prompt + runtime DLP scan
Action Limits	Never delete records without human approval; never make purchases > \$X without authorization	Policy engine hard constraints
Communication	Never impersonate humans; always identify as AI when asked; never send on behalf of user without explicit permission	Output filter + system prompt
Scope Adherence	Stay within task scope; do not pursue side goals; do not accumulate resources beyond task needs	Goal consistency monitor
Transparency	Always explain reasoning when asked; never deceive principals about actions taken	Audit trail + honesty check
Safety Priority	Prioritize safety over task completion; halt and escalate if safety constraint conflicts with task	Priority hierarchy in system prompt

10.5 Supervisor Agent Architecture

10.5.1 Using AI to Oversee AI

Supervisor agents provide scalable oversight by using AI to review AI actions at a semantic level beyond what rule-based policy engines can achieve. The supervisor is a separate model (different architecture or fine-tune than the task agent) that receives: the agent's reasoning trace, proposed action, and task context, and returns a safety classification and risk assessment.

■ **Critical Requirement:** The supervisor agent must be architecturally isolated from the task agent. If both use the same base model, a prompt injection targeting the base model may defeat both simultaneously (correlated failure). Use different model families, different fine-tuning histories, or different providers for supervisor and task agent.

Supervisor Function	Input	Output	Implementation
Reasoning Sanity Check	Agent reasoning trace	Anomaly flag + explanation	Dedicated review model
Plan Consistency Check	Plan + stated goal	Consistency score 0-100	Goal consistency classifier
Output Safety Review	Agent output + context	Safety classification	Constitutional AI model
Behavioral Pattern Review	Action sequence history	Pattern anomaly score	Sequence analysis model
Escalation Recommendation	Risk signals + context	Escalate / proceed / block	Risk aggregation model